Government Bye-Laws, Guidelines & Existing Systems for 3D Cadastre

1: The Regulatory Landscape & Statutory Hierarchy

- **Context:** Land and property administration in India operates across a three-tier regulatory framework. To scale a 3D cadastre, the prototype must respect this chain of authority rather than bypass it.
- **National Framework:** Governed by Acts like the **Real Estate (Regulation and Development) Act (RERA), 2016** (which legally standardizes carpet area definitions) and the **Digital Personal Data Protection (DPDP) Act, 2023** (which dictates privacy and data handling).
- **State & Municipal Framework:** State-level planning instruments like **Maharashtra's Unified Development Control and Promotion Regulations (UDCPR)** govern local development rules.
- **The Core Problem:** Traditional governance is built on flat, 2D boundaries (such as 7/12 extracts and property cards), creating massive blind spots for multi-level ownership, basements, and airspace.

2: Planning Rules & Development Controls (FSI, Height & Setbacks)

- **Floor Space Index (FSI) / FAR:** FSI is defined as the ratio of total built-up floor area to plot area. It is *not* a universal 3D volumetric maximum. The system must compute FSI dynamically using local sanctioned-plan inputs.
- **Building Heights & Margins (UDCPR Standards):**
    - Low-rise vs. high-rise classifications determine mandatory side, rear, and front setbacks.
    - For instance, buildings exceeding specific height thresholds trigger mandatory Fire Department (CFO) NOCs and wider access roads.
- **Prototype Implementation:** Our 3D engine configures local planning rule engines to validate volumetric models against statutory limits as a spatial diagnostic tool, leaving final legal approvals to human authorities.

3: Existing Systems & The Tech Gap

- **Current Municipal Portals:** Local bodies (such as the Pune Municipal Corporation) use automated building plan scrutiny systems like **AutoDCR / PreDCR** alongside Single Window Clearance (SWC) portals for digital approvals.
- **The Technological Limitation:**
    - Existing systems operate strictly in **2D CAD layers and flat database structures**.

- They struggle to seamlessly validate overlapping vertical boundaries (e.g., stacked commercial ownership, basement parking bays, and public utility/metro corridors running beneath buildings).

- **The Gap We Are Solving:** Moving from legacy 2D blueprints to a searchable, multi-level 3D spatial index that links directly to authoritative parcel IDs (ULPIN) without rewriting legal titles.

4: Compliance & Risk Mitigation (Jury Defense)

- **Ownership vs. Modeling:** The system *visualizes and manages* spatial records; it does not create legal title merely by generating a 3D object or string identifier.
- **Clash Detection Logic:** Using spatial predicates (like 3D geometric intersections) only flags physical overlaps or potential encroachments; it acts as an **evidence-support signal** for registrars, not an automated court verdict.
- **Privacy & DPDP Compliance:** Moving beyond basic data masking by integrating multi-layered safeguards—including role-based access control, data minimization, secure audit logging, and strict separation of public spatial maps from personal identity data.

Statutory Backbone: RERA 2016 & Real Estate Compliance

- **Legal Definition of Carpet Area:** Under Section 2(k) of the Act, "carpet area" is legally defined as the net usable floor area of an apartment (excluding external walls, service shafts, and exclusive open terraces or balconies).
- **Prototype Alignment:** Our 3D cadastre prototype respects this statutory definition by maintaining distinct boundary layers for carpet areas rather than relying on arbitrary 3D volumetric extrusions.
- **Sanctioned Plans & Transparency (Section 4 & 11):** Promoters are legally required to upload approved layout plans, stage-wise schedules, and fire-fighting facility details onto the regulatory web portal for public viewing.
- **Structural Accountability (Section 14):** Promoters remain legally responsible for rectifying any structural defects or quality issues brought to light by allottees within a **5-year window** from handing over possession.
- **The Tech-to-Law Bridge:** While RERA mandates digital project disclosures and transparent tracking, legacy portals rely on flat documents. Our 3D cadastre elevates these mandatory disclosures into an interactive, multi-level spatial index.

- **Definition of High-Rise Buildings:** Buildings higher than 15 meters without stilts and above 17.5 meters with stilts are officially classified as high-rise developments.
- **Mandatory Fire Safety and Access:** High-rise structures and special occupancies require dedicated perimeter open spaces (at least 6 meters wide up to 40 meters height, and 9 meters beyond that) capable of supporting heavy fire tenders weighing up to 45 tonnes.
- **Structural Safety & Review Panels:** Structural designs must comply with National Building Code (NBC) and Bureau of Indian Standards (BIS) codes, with mandatory third-party proof-checking by a Structural Design Review Panel (SDRP) for buildings taller than seven storeys.
- **Sustainability & Rainwater Harvesting (RWH):** RWH is compulsory for all new building plots of 100 square meters or more, and wastewater recycling systems are mandatory for buildings generating a minimum discharge of 10,000 liters per day.

Development Control Framework & Approvals (UDCPR)

- **Unified State Standard:** Governed by Maharashtra's **UDCPR** framework, standardizing zoning, setbacks, and FSI.
- **High-Rise Definition:** Buildings with a height of **24 meters or more** are classified as high-rise developments.
- **Environmental & Site Constraints:** Strict prohibitions against building within **6 meters of minor water courses (nallahs)** and **15 meters of major water courses (rivers)**, alongside strict height and safety regulations inside Red and Blue flood lines.

Parking Standards & Dimensions (Chapter 8 / MBBL)

- **Vehicle Parking Bay Sizes:**
    - **Large Car:** $5.0 \text{ m} \times 2.5 \text{ m}$ (up to 50% can be small car size at $4.5 \text{ m} \times 2.3 \text{ m}$).
    - **Two-Wheeler (Scooter/Motorcycle):** $1.0 \text{ m} \times 2.0 \text{ m}$.
    - **Transport Vehicle/Ambulance:** $3.75 \text{ m} \times 7.5 \text{ m}$.
- **Maneuvering & Ramps:** Minimum width for driveways is **3.0 meters** for motor vehicles. Basements/upper floors require proper ramps with slopes not steeper than **1:8 or 1:10**.
- **Loading & Unloading:** Commercial, mercantile, and industrial buildings must provide explicit loading/unloading bays ($3.75 \text{ m} \times 7.5 \text{ m}$) at a rate of 1 space per 1,000 sq.m. of floor area.

Building Space & Component Requirements (Chapter 9)

- **Habitable Rooms:** Minimum size is **9.5 sq.m.** with a minimum clear width of **2.4 meters**. Minimum clear ceiling height is **2.75 meters**.
- **Kitchens:** Minimum size of **5.0 sq.m.** (width 1.8m) if separate, or **7.5 sq.m.** (width 2.1m) if combined with a dining area.
- **Bathrooms & WCs:** Minimum bathroom size is **1.5 sq.m.** (1.0 m width); standalone WC is **1.0 sq.m.** (0.9 m width); combined bath/WC is **2.4 sq.m.**.
- **Staircases & Corridors:** Minimum clear width for residential staircases is **1.0 meter** (higher for commercial/assembly up to $1.5 \text{ m} - 2.0 \text{ m}$), with maximum riser height restricted to **190 mm** for residential and **150 mm** for other buildings.

1. The Core Concept: Moving from 2D to 3D Geo-Referenced Data

- **What DILRMP Does:** The DILRMP framework focuses on digitizing land records, integrating textual Records of Rights (RoR) with spatial/cadastral maps, and assigning unique identifiers like the **ULPIN (Unique Land Parcel Identification Number)**—also known as *Bhu-Aadhaar*.
- **How to Pitch It in the PPT:** Your 3D cadastre project takes the national ULPIN concept and extends it vertically. While DILRMP successfully maps ground-level footprints (2D parcels), modern multi-story urban developments need a **3D Vertical ULPIN schema** to index stacked properties, basements, and airspace.

2. The Shift from Presumptive to Conclusive Titling

- **The National Goal:** A major objective of the DILRMP under DoLR is to transition India from a *presumptive titling system* (where registration of deeds only proves a transaction took place, not absolute ownership) to a **conclusive land-titling system with title guarantees**.
- **How to Pitch It in the PPT:** Emphasize that your 3D prototype serves as an advanced **spatial decision-support and visualization engine**. It helps bridge the gap toward accurate multi-level land administration, supporting future title guarantees by resolving vertical overlaps and volumetric boundaries that flat 2D maps cannot represent.

3. Solving Urban Land Disputes & Integration Gaps

- **The Problem Highlighted by DoLR:** Legacy land administration suffers from discrepancies between textual records and physical maps, resulting in massive urban litigation. Furthermore, traditional systems completely lack mechanisms to handle complex multi-story ownership, underground utilities, or overlapping air rights.
- **How to Pitch It in the PPT:** Position your 3D cadastre as the architectural solution to the "vertical blind spot" in modern land governance, aligning with the government's vision for seamless, transparent, and technology-driven land information systems (ILIMS).

Prototype Solution & Risk Management

Headline: Bridging Tech Verification with Statutory Authority

- **Clash Detection Logic:**
  - *What it does:* Uses spatial tools (like 3D intersections via PostGIS) to flag physical overlaps or geometric cross-overs.
  - *The Legal Boundary:* This acts strictly as a **decision-support signal** for the registrar, not an automated court verdict on an encroachment.
- **Subsurface & Airspace Handling:**
  - *System Design:* Treats underground utility corridors and upper airspace volumes as distinct spatial objects linked directly to title documents.
  - *Statutory Foundation:* Governed conceptually by the principles of the **Indian Easements Act, 1882**, which handles rights to light, air, and property access.
- **The Jury Takeaway:**
  - The platform respects the chain of command. Automation assists the human registrar in catching spatial conflicts, but **the State holds the ultimate legal authority** to adjudicate ownership.

Official References to Cite:

- *Indian Easements Act, 1882* (for air, light, and property access rights)
- *PostGIS Spatial Predicates* (for computational geometry validation)

Tip for the Pitch:

If the jury asks: *"Can your software automatically prove someone is illegally encroaching on underground space?"* Your teammate can confidently reply using this slide's logic: "No. PostGIS flags the geometric intersection, but our system leaves the legal determination of an easement or ownership violation to human authorities reviewing the official title deeds."

Hackathon Presentation Brief: 3D Vertical Land & Property Cadastre

Slide 1: The Regulatory Landscape & Statutory Hierarchy

- **National Consumer Protection:** Governed by the **Real Estate (Regulation and Development) Act (RERA), 2016**, which legally standardizes *carpet area* definitions (Section 2k) and mandates promoter financial transparency (putting 70% of collections into a separate construction account).
- **Data Privacy:** Aligned with the **Digital Personal Data Protection (DPDP) Act, 2023**, enforcing data minimization, encryption, role-based access, and the strict separation of public spatial maps from private citizen identity data.
- **State Planning Framework:** Governed by Maharashtra's **Unified Development Control and Promotion Regulations (UDCPR)** framework, which sets standardized building, zoning, and height parameters.
- **The Core Problem:** Legacy land systems rely on flat, 2D boundaries (such as 2D property cards and paper titles), creating a severe "vertical blind spot" for multi-level ownership, basements, and airspace.

Slide 2: Planning Rules, FSI & Site Eligibility

- **Floor Space Index (FSI) Mechanics:** FSI is defined as the ratio of total built-up area to plot area. It is *not* a universal 3D volumetric maximum. Our engine computes FSI dynamically using local sanctioned-plan inputs.
- **Site Eligibility & Water Protection (UDCPR Chapter 3):** Strict environmental constraints dictate that no construction is allowed within **6 meters of minor water courses (nallahs)** and **15 meters of major water courses (rivers)**, alongside strict height limits within Blue and Red flood lines.
- **High-Rise Classifications:** Buildings exceeding **24 meters** in height are formally classified as high-rise structures under state standards, triggering mandatory high-rise provisions.

Slide 3: Micro-Specs: Parking & Building Spatial Requirements

- **Vehicle Parking Standards:**
  - Large Car Bays: $5.0\text{ m} \times 2.5\text{ m}$(up to 50% can be small car bays at $4.5\text{ m} \times 2.3\text{ m}$).
  - Two-Wheelers: $1.0\text{ m} \times 2.0\text{ m}$.
  - Loading/Unloading Bays: $3.75\text{ m} \times 7.5\text{ m}$for commercial/industrial blocks.

- **Ramps & Maneuvering:** Driveways require a minimum width of **3.0 meters**, with ramp gradients restricted to **1:8 or 1:10** for safe vehicular and fire-tender movement.
- **Room & Structural Dimensions:** Habitable rooms require a minimum area of **9.5 sq.m.** (minimum width of 2.4m) and a clear ceiling height of **2.75 meters**. Residential staircases must maintain a minimum clear width of **1.0 meter** with maximum risers capped at **190 mm**.

Slide 4: Prototype Solution, Clash Detection & Risk Management

- **Clash Detection Logic:**
    - *Tech Used:* Spatial algorithms (like 3D intersections via PostGIS) to flag physical overlaps.
    - *Legal Boundary:* Acts strictly as a **decision-support signal** for registrars, not an automated court verdict on an encroachment.
- **Subsurface & Airspace Handling:** Treats underground utility corridors and upper airspace volumes as distinct spatial objects linked to title documents, conceptually anchored by the **Indian Easements Act, 1882**.
- **The Jury Takeaway:** The platform respects the chain of command—automation assists the human registrar in detecting conflicts, but **the State retains absolute legal authority** to issue titles.

Standardized 3D ULPIN String Specification & Examples

- **Standardized Format Syntax**: [14-Digit Base ULPIN]-[Block]-[Floor]-[Unit]
- **Field Semantics & Breakdown**:
    - **Base ULPIN**: Authoritative 14-digit base parcel identifier issued by the land-record system (e.g., 29-XX-1234567890).
    - **Block / Wing**: Building or wing identifier within the project dataset (e.g., WB).
    - **Floor / Level**: Vertical level convention (positive numbers for floors, negative numbers like -B01, -B02 for basements, or -SUB01 for subsurface utility corridors).
    - **Unit / Right**: Specific spatial object identifier (e.g., 402, P-B17 for parking bays, or AIR-MAX for upper spatial-right volumes).

Real-World Mockup Examples

- **Example 1: Standard Residential Flat (Flat 402, Wing B)**
    - **String Breakdown**: ULPIN3D: 29-XX-1234567890:WB:L04:402:UNIT
    - **Interpretation**: Base parcel reference + Wing B + Level 04 + Unit 402. *Note for the Jury:* This identifier models the 3D spatial relationship, but legal ownership remains tied to authoritative title and registration records.
- **Example 2: Commercial Basement Parking Bay (Basement-2, Parking Bay 17)**
    - **String Breakdown**: ULPIN3D: 29-XX-1234567890:WA:-B02:P-B17:UNIT
    - **Interpretation**: Base parcel reference + Wing A + Basement 2 + Parking Bay 17. *Note for the Jury:* Subsurface assets like parking bays or utility corridors are modeled as distinct spatial objects linked to governing title deeds and sanctioned plans.

1. Mobile Field Surveyor Survey App (PWA)

- **What it is:** Think of this as a special mobile app for surveyors who go out into the field to inspect buildings. "PWA" stands for Progressive Web App, meaning it works right in the mobile browser like an app without needing a heavy download.
- **What it does:** When a surveyor visits a building, the app records critical details: the exact GPS location (latitude and longitude), how many floors the building has, the specific unit number, and floor area measurements. They can also snap pictures of the exterior structure, tax receipts, and registry deeds to upload right from their phone.
- **How to talk about it:** "Our Mobile Surveyor App acts as the data-collection frontline. Instead of surveyors carrying around messy paper clipboards, they use a mobile-friendly PWA to capture real-time GPS coordinates, floor counts, and photos, pushing verified data straight back into our central 3D cadastre database."

2. The Registrar Compliance Control Panel

- **What it is:** This is the dashboard used back at the office by government officials (the registrars). Think of it like a smart control room screen that flags problems automatically.
- **What it does:** The system uses 3D spatial calculations to check if a new building or floor design crosses legal lines. It flags two main types of alerts:
  - **RED (Encroachment/Spatial Clash):** Triggered if a 3D building model overlaps with a protected space, a neighboring property, or underground infrastructure.
  - **YELLOW (FSI/Planning Breach):** Triggered if calculations show the building breaks local Floor Space Index (FSI) or planning limits.
  - From this dashboard, the registrar can click buttons to **Approve**, **Flag a Dispute & Hold**, or **Send Back for Resurvey**.
- **How to talk about it:** "The Registrar Control Panel is the command center where automation meets human authority. If a building plan violates FSI rules or geometrically overlaps with an underground metro line or neighbor's property, our 3D engine flags it as a RED or YELLOW alert. The human registrar reviews the evidence and clicks 'Approve' or 'Send Back for Resurvey', ensuring technology assists decision-making without replacing legal accountability."

3. The 1-Click "Vertical Property Card" (PDF)

- **What it is:** This is the modern, 3D equivalent of the traditional flat paper property card. With a single click, it generates a downloadable, official-looking PDF document for a specific 3D unit or property volume.
- **What's inside it:**
    - A **QR Code** that links securely to the digital record (without showing sensitive personal names for privacy).
    - A **Static 3D Isometric Snapshot** of that exact property volume.
    - **Volumetric Dimensions & Net Volume ($X \times Y \times Z$ in meters and m^3).**
    - A **Landowner Token** and a **Registrar Digital Signature placeholder**.
- **How to talk about it:** "Just like land titles have 2D property cards, our 1-Click Vertical Property Card generates a comprehensive PDF certificate for 3D multi-level properties. It includes a secure QR code, an isometric 3D snapshot of the exact volumetric space (m^3), and digital signature placeholders, giving owners a transparent, modern proof of their vertical property boundaries."